SAURABH UPADHYAY

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(Stamp 2, eligible for Stamp 1G)

PROFESSIONAL SUMMARY

MSc Business Analytics student at University College Cork with over three years as a software test engineer at Sage, now moving into business analysis. Spent that time turning customer issues and feature requests into acceptance criteria, tracing recurring defects back to root causes and verifying payroll data before release. Led a stakeholder research project that mapped delivery bottlenecks across the SDLC and presented the findings to senior leadership.

SKILLS

Data and Analytics: Python (pandas, NumPy, scikit-learn, XGBoost, statsmodels), SQL, time series forecasting (Holt-Winters, SARIMA, ARIMA, logistic growth), classification and model comparison, model interpretability (SHAP, LIME), linear and integer programming (PuLP), Monte Carlo simulation, K-means clustering, Advanced Excel (PivotTables, Power Query, lookups)

Business Analysis: Requirements gathering, user stories and acceptance criteria, stakeholder mapping and management, process mapping, gap analysis, root cause analysis, user research and journey mapping, UAT, Agile and Scrum, SDLC

Technical: REST API integration and testing, Streamlit, Plotly, NetworkX, test automation with Cypress, CI pipelines (Azure DevOps, TeamCity), JMeter, HTML, CSS, JavaScript

Tools: Git and GitHub (branch protection, PR review), Jupyter, Figma, Jira, Confluence, Postman

EDUCATION

University College Cork **Sep 2025 to Present**
MSc Business Analytics

Lovely Professional University, Jalandhar **Aug 2017 to May 2021**
Bachelor of Technology, Computer Science

PROFESSIONAL EXPERIENCE

Sage **Oct 2021 to Jun 2025**

Mohali, India (Hybrid) **Products:** Sage Business Cloud Payroll and HR | AutoEntry by Sage

Software Test Engineer *Oct 2023 to Jun 2025*

- Worked with product owners and developers across the Payroll and HR release cycle, translating customer issues and feature requests into acceptance criteria the team could build and test against.
- Investigated high priority customer incidents and traced recurring defects back to root causes, then fed those findings into sprint planning and prioritisation so the same issues stopped resurfacing.
- Built front end and back end automation in Cypress from scratch and worked with a developer to wire the suite into CI, with Azure DevOps triggering runs on TeamCity, replacing manual regression passes and cutting the effort required on every release.
- Verified data integrity across APIs and back end services, reconciling expected against actual results on payroll calculations before they reached customers.
- Ran release and post deployment verification and supported live incident management, coordinating between engineering and customer support on escalations.

Associate Software Test Engineer *Oct 2021 to Oct 2023*

- Led Project Process Improvement as project manager during the Graduate Bootcamp. Interviewed stakeholders across the delivery function to map how work actually moved through the SDLC, identified where it was getting stuck and presented recommendations to senior leadership, with a projected reduction in turnaround time of around 20 percent.
  Awarded Boldest Team, December 2022.
- Worked with business stakeholders on AutoEntry features, pushing for clarity on ambiguous requirements before development began, which reduced rework later in the cycle.
- Completed SAGE IGNITE, a two year graduate programme covering commercial awareness, stakeholder communication and delivery methodology.

PROJECTS

National Transport Decarbonisation Dashboard for Ireland, University College Cork **Mar 2026 to Jul 2026**

- Built a five page analytics dashboard answering whether Ireland is on track for its 2030 transport decarbonisation targets, replacing a manual process where analysts stitched CSO spreadsheets together and produced conflicting narratives from the same data.
- Found that EV uptake plateaus 18.4 percentage points below the Climate Action Plan 2030 target and translated that gap into 1,860 additional EV and hybrid registrations needed per month, giving planners a number to plan against rather than a trend line.
- Modelled nine CSO PxStat and data.gov.ie datasets (roughly 300,000 rows, 2015 to 2026) into three composite KPIs and forecasts using Python, Holt-Winters, SARIMA, logistic growth and K-means clustering, with MAPE reported per model.
- Group of seven. Owned the GitHub repository and PR review process, built the Streamlit application and the forecasting notebooks, and caught a dataset scope mismatch and a model specification error during review that would have skewed the published projections.
- **Dashboard:** https://ie-transports-decarbonization-tracker.streamlit.app
  Code: https://github.com/srb1024/IE_TRANSPORTs_DECARBONISATION_TRACKER

Global Supply Chain Optimisation for a Perfume Manufacturer, University College Cork **Feb 2026 to Apr 2026**

- Quantified a 35,000 unit monthly capacity shortfall as EUR 37.5M a year in unmet demand penalties and found 20,000 units of idle capacity that could be activated at a net saving of EUR 150 to 250 per unit, giving the client an ROI benchmark for expansion.
- Built a mixed integer linear programme in Python and PuLP across seven decision variable classes, after reshaping the raw supplier, plant, workforce and demand tables into the parameters the model needed.
- Turned the model output into a recommendation the client could act on, arguing for activating idle capacity before committing to expansion, with the ROI benchmark to support it.
- **Approach:** https://srb1024.github.io/projects/supply-chain

Customer Churn Prediction for a Telecoms Provider, University College Cork **Oct 2025 to Dec 2025**

- Engineered the modelling dataset for a churn study on 50,000 customer records, standardising inconsistent categorical entries into 114 encoded predictors and adding two behavioural features covering spend per month of tenure and recent engagement against historic activity.
- Compared six classifiers tuned on stratified cross-validation and selected XGBoost at 84 percent accuracy and 86 percent precision, then applied SHAP and LIME so each flagged customer arrives with the reasons attached rather than a bare score.
- Identified payment delinquency, short tenure and competitor pressure as the drivers worth acting on rather than the demographics the business had been segmenting by and recommended continuous CRM scoring with auto-pay incentives and first-six-month intervention.
- **Approach:** https://srb1024.github.io/projects/churn

Urban Moonlight, Smart Streetlighting Concept, University College Cork **Sep 2025 to Dec 2025**

- Reframed a smart streetlighting brief for Cork City after stakeholder and empathy research found that fear of falling, isolation and unfamiliarity with public technology kept elderly and disabled residents indoors after dark, not poor visibility.
- Translated those findings into design requirements for a retrofit rather than a replacement, covering motion-activated lighting, an emergency call function, voice and braille accessibility panels and charging points.
- Ran ideation and built the working prototype in Figma Make, then defined the usability and pilot testing framework used to validate it. **Prototype:** https://earth-folk-30984861.figma.site

CERTIFICATIONS

Pluralsight: Introduction to the BABOK Guide and Business Analysis Key Terms, *Jun 2022* | Business Analysis: Skills and Competencies of Effective Business Analysts, *Jun 2022* | Agile Fundamentals, *Feb 2022* | Introduction to SQL, *Feb 2022*

REFERENCES

Available on request.